A393171

A Catalan substitution congruence modulo eight

Dev Goyal

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AI disclosure. ChatGPT (OpenAI) generated the mathematical arguments, Lean proofs, and this manuscript in response to Dev Goyal's requests. Dev Goyal is the sole human author. The source sequence and conjecture or observation are credited to Paul D. Hanna. **Independent human review is pending.**

Abstract

For $C = x + C^2$, the unique normalized integral solution of $B = x + B(C)^2$ has residue four modulo eight precisely at indices 2^k or $2^k + 1$, $k \geq 1$, above degree two. We prove the prerequisite modulus-four result and an exact integral lift, so this manuscript is independent of the companion paper.

1 Source, notation, and scope

The source is Paul D. Hanna's OEIS entry A393171 [1], retrieved on 24 October 2026. Its initial sequence index is 1. We also retain the formal zero constant coefficient at index zero.

All series are formal. The set $\mathbb{N}$ includes zero, and $[x^n]U$ denotes a coefficient. A normalized series is $x + O(x^2)$; $O(x^r)$ means divisibility by x^r, over any coefficient ring used below. We write $U \circ V = U(V(x))$ only when $V(0) = 0$. Compositional inversion, denoted $\mathrm{Rev}(U)$ or $U^{\langle -1 \rangle}$, is distinct from multiplicative inversion of a unit. Congruences are coefficientwise; ordinary products and compositional iterates are distinct.

2 Formal algebra used in the proof

Lemma 2.1 (Triangular construction)**.** *Let K be a commutative ring. Suppose Φ preserves normalized series and improves agreement by one degree:*

$$G - H = O(x^r) \quad \Longrightarrow \quad \Phi(G) - \Phi(H) = O(x^{r+1}) \qquad (r \geq 2).$$

Then Φ has a unique normalized fixed point. The same conclusion holds for a family of series if the degree-n equations, at every family index, determine the degree-n coefficients from strictly lower degrees.

Proof. The degree-n coefficient of $\Phi(G)$ depends only on coefficients of G below degree n. Fix the constant and linear coefficients, and then define every later coefficient successively. This gives a fixed point; the same induction proves uniqueness. For a family, perform the induction on degree simultaneously at every family index. The same induction applies with only the constant coefficient prescribed, provided agreement is improved starting at degree one. It also applies to an operator on all series when agreement is improved starting at degree zero. For the

operators used in this paper, the recurrence requires only ring operations, so it also constructs integral solutions when $K = \mathbb{Z}$. Equivalently, iterating Φ from any normalized initial series fixes one further coefficient at each step. □

Lemma 2.2 (Inversion and transport)**.** *A normalized series over any commutative ring has a unique normalized two-sided compositional inverse. A constant-one series has a unique multiplicative inverse with constant coefficient one. These inverses and valid composition commute with coefficient ring homomorphisms. Normalized compositional inversion preserves agreement modulo x^r.*

Proof. For normalized G, the degree-n equation in $G(H) = x$ determines h_n with multiplier one from earlier coefficients. Construct a right inverse J of H. Associativity gives $G = G \circ (H \circ J) = (G \circ H) \circ J = J$, so $H \circ G = x$ as well. The same associativity argument proves uniqueness. Multiplicative inversion follows similarly from the coefficient recurrence for $UV = 1$.

Each coefficient of a valid composition is a finite polynomial in the coefficients of its arguments. Ring homomorphisms preserve these polynomials and the defining equations, so uniqueness gives transport of inversion. If $G - H = O(x^r)$, composition with any normalized series preserves that agreement. Composing with the respective inverses then shows $\mathrm{Rev}(G) - \mathrm{Rev}(H) = O(x^r)$. □

Lemma 2.3 (Binary series and doubling)**.** *There is a unique normalized integral series $C = x + C^2$. Its reduction modulo two is*

$$P(x) = \sum_{k \geq 0} x^{2^k}.$$

For series U, V over $\mathbb{Z}/4\mathbb{Z}$, the equality $2U = 2V$ is equivalent to equality of their reductions modulo two.

Proof. The operator $G \mapsto x + G^2$ improves agreement because $G^2 - H^2 = (G - H)(G + H)$ and normalized $G + H$ has zero constant coefficient. In characteristic two, $U(x)^2 = U(x^2)$ for every formal series. Thus the normalized equation $U = x + U^2$ forces the coefficient at one to be one, every higher odd-index coefficient to be zero, and the coefficient at $2m$ to equal the coefficient at m. This gives exactly P.

For scalar residues modulo four, $2u = 2v$ holds precisely when u and v have the same parity. Apply this statement coefficientwise. In particular, this argument does not cancel a factor of two in a ring in which two is a zero divisor. □

3 A prerequisite congruence and the modulus-eight lift

Let $C = x + C^2$ and put $N = x - x^2$. Then $N \circ C = x$, and Lemma 2.2 also gives $C \circ N = x$. We use A for A393172 and B for A393171 throughout this section, to keep the two source series distinct [2, 1].

Lemma 3.1 (A useful square equation)**.** *For normalized $T, P \in K[[x]]$, there is a unique normalized solution of*

$$G = T + (G \circ P)^2.$$

The construction commutes with coefficient ring homomorphisms.

Proof. If $G - H = O(x^r)$, then $(G - H) \circ P = O(x^r)$. The sum $G \circ P + H \circ P$ has zero constant coefficient, so the difference of the two squares is $O(x^{r+1})$. Normalization is preserved. Lemma 2.1 gives existence and uniqueness; mapping the equation and using uniqueness gives transport. □

Theorem 3.2 (A393172 modulo four)**.** *The unique normalized integral solution of*

$$A \circ N = x + A^2 \tag{1}$$

has, for every $n > 1$,

$$[x^n]A \equiv \begin{cases} 2 \pmod 4, & n = 2^k \text{ for some } k \geq 0, \\ 0 \pmod 4, & \text{otherwise.} \end{cases}$$

Proof. Composing (1) with C gives $A = C + (A \circ C)^2$. Conversely, composing this equation with N gives (1). Lemma 3.1 therefore proves integral existence and uniqueness for the literal source equation.

Over $\mathbb{Z}/4\mathbb{Z}$, the normalized candidate $D = 2C - x$ satisfies

$$D \circ N = 2x - N = x + x^2, \qquad D^2 = x^2 + 4(C^2 - xC) = x^2.$$

Thus D solves (1), and uniqueness gives $A = 2C - x$ modulo four. For $n > 1$, the coefficient of x contributes nothing and Lemma 2.3 shows that twice the coefficient of C is two precisely at powers of two, and zero elsewhere. □

Theorem 3.3 (A393171 modulo eight)**.** *There is a unique normalized integral series*

$$B = x + (B \circ C)^2.$$

For every $n > 2$,

$$[x^n]B \equiv \begin{cases} 4 \pmod 8, & n = 2^k \text{ or } 2^k + 1 \text{ for some } k \geq 1, \\ 0 \pmod 8, & \text{otherwise.} \end{cases}$$

Moreover $B = x + A^2 = A \circ N$ and $B \circ C = A$.

Proof. Existence and uniqueness follow from Lemma 3.1 with $T = x$, $P = C$. Define $B = x + A^2$. The fixed point equation for A gives $B \circ C = C + (A \circ C)^2 = A$, so this B satisfies the required source equation. Equation (1) gives $B = A \circ N$; it also implies $B \circ N = N + B^2$.

Theorem 3.2 provides an integral series K with $A = 2C - x + 4K$. Squaring over the integers and reducing modulo eight gives

$$\begin{aligned} B &\equiv x + (2C - x)^2 \\ &= x + x^2 + 4(C^2 - xC) \\ &= x + x^2 + 4(C - x - xC) \pmod 8. \end{aligned} \tag{2}$$

For $n > 2$, the relevant coefficient divided by four has parity $[x^n]C + [x^{n-1}]C$. By Lemma 2.3, it is one precisely when one of $n, n-1$ is a power of two. They cannot both be powers of two for $n > 2$: consecutive powers of two can only be one and two. The exact exceptional indices are therefore the two families stated above. □

The passage to (2) first establishes exact divisibility by four in the integer series; it does not divide by four in $\mathbb{Z}/8\mathbb{Z}$. The canonical constructions, both source equivalences, the identities relating A and B, and both full residue classifications are formalized in `Round4.CatalanComposition`. The general contraction construction is in `Round4.SquareComposition`.

4 Separate Lean proof and checked scope

The entry file is `lean/Entries/A393171.lean`. Its explicit theorem statements and proof terms use the namespace `FinalCollection.A393171`. The complete imported implementations and their shared mathematical dependencies are included in the collection. The following local theorem names give the proof map:

- `second_existence`.
- `second_named_existence`.
- `second_definition`.
- `second_residue`.
- `second_arbitrary_source`.
- `link_to_first_sequence`.
- `link_composed_catalan`.
- `complete_series_mod_eight`.
- `divisibility_complement`.

The main implementation modules are

`Round4.CatalanComposition`

`Round4.SquareComposition`

The catalogue records the full original endpoint names and the new entry theorems. The project pins Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The collection report records source recompilation and transitive axiom checks. Accepted axioms are propositional extensionality, classical choice, and quotient soundness. All index conclusions above are unbounded; exact finite coefficient checks in the package provide separate corroboration.

Limits and status. The exact integral decomposition $A = 2C - x + 4K$ is proved before reducing modulo eight. The manuscript consolidates completed results; it does not assert global novelty, exclusive priority, peer review, OEIS acceptance, or a new submission. Standard algebraic ingredients are not claimed as new discoveries.

References

[1] Paul D. Hanna, *OEIS A393171*, 4 February 2026. `https://oeis.org/A393171`. Retrieved 24 October 2026.

[2] Paul D. Hanna, *OEIS A393172*, 4 February 2026. `https://oeis.org/A393172`. Retrieved 24 October 2026.